



Heimkommen

Will I get home tonight?

Regional journey planning with real delay risk for the Schwarzwald-Baar-Heuberg region

Project report · September 2026

github.com/Niran1202/heimkommen ·
github.com/Niran1202/heimkommen-desktop

1. Summary

Most journey planners assume every connection works. Heimkommen instead estimates **how likely** each connection is to work, based on six months of real train delays, and answers the question a traveller actually has late in the evening: *will I get home tonight, and which train should I take?*

For every journey the app shows the probability that all connections work, the probability of arriving at most five minutes late, the probability of getting home at all tonight, the weakest connection with a Plan B, and the latest departure that is still safe.

In a replay of 891 real evening journeys from July and August 2026 (months the model never saw), Heimkommen's probabilities were clearly better than both a timetable-only planner and a historical-median approach on every measure (section 7).

The project is available in three forms built from the same code:

Form	Where	Notes
Web application	github.com/Niran1202/heimkommen	Accounts, saved trips, nightly accuracy tracking, Docker deployment
Windows desktop app	github.com/Niran1202/heimkommen-desktop (release v1.0.0)	Installer or portable zip, no login, everything bundled
Local development	Developer PC	SQLite database, runs without Docker

2. How it works

Step	What happens
1. Routing	A RAPTOR route planner finds journeys in the official NVBW timetable.
2. Delay prediction	A machine-learning model predicts, for each train, the range of delays at the boarding and alighting station, plus its chance of being cancelled.
3. Simulation	2,000 possible evenings are played out per journey: sample delays, check every connection, switch to the next train if one is missed, mark the run “stranded” if nothing is left that night.
4. Answer	The share of successful runs becomes the probabilities shown to the user.

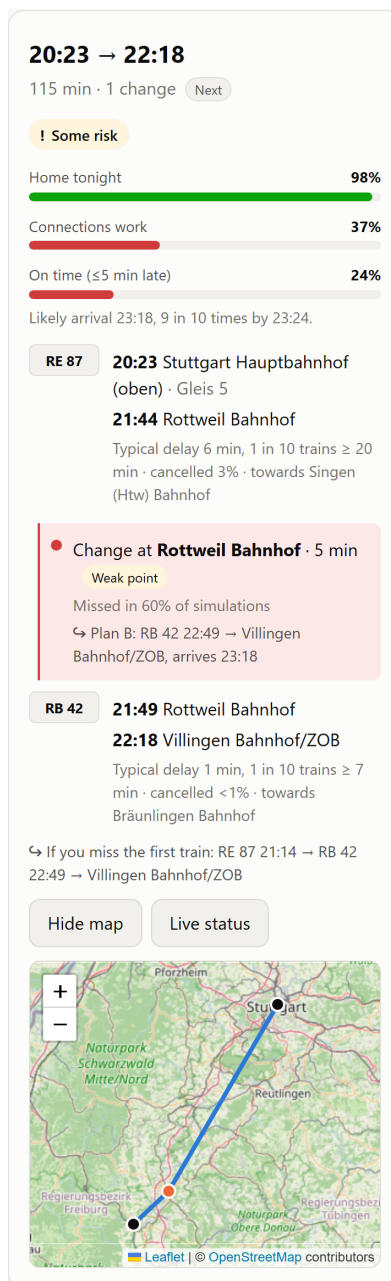


Figure 1 – A journey on a phone: risk per connection, the weak point (5-minute change in Rottweil), Plan B and the map.

3. Data sources

Data	Source and licence	What was used	Used for
Timetable	NVBW “bwgesamt” GTFS feed, version 20260913 (CC BY 4.0)	705 MB ZIP, valid 3 May – 12 Dec 2026	Routing
Historical delays	piebro/deutsche-bahn-data on Hugging Face (CC BY 4.0, data by Deutsche Bahn)	6 monthly files, March–August 2026, ~600 MB each (all of Germany)	Training and evaluation
Yesterday's train times	Same dataset, raw daily API responses	e.g. 22 Sep 2026: 112,252 stop events	Nightly accuracy check
Live delays	DB Timetables API (DB API Marketplace, free plan)	Only on request; cached 60 s, max. 30 calls/min	Live status
Map tiles	OpenStreetMap (ODbL)	Only when a map is opened	Journey map

Imported timetable

Rail was kept for all of Baden-Württemberg (so long regional trips can be planned); buses only in the three region districts. Result: 527 routes, 65,918 trips, 1,122,593 stop times, 8,549 stops, 8,806 service calendars, 337,769 calendar exceptions and 15,966 transfer rules, imported in 80 seconds.

Delay history

After filtering to the matched stations, about 3 million stop records per month remain (roughly 18 million in total). Rail-replacement bus rows and unscheduled extra stops were removed.

4. Linking the two datasets

The timetable and the delay data name stations differently ("Villingen Bahnhof/ZOB" vs. "Villingen (Schwarzw)"). Stations were therefore matched by **timetable coincidence**: if the same train number stops at the same minute in both datasets, that is a vote for the pair.

- Input: four days in August 2026 – 450,666 timetable events and 1,838,273 delay-data events.
- Votes are counted per distinct train, so a single train number that collides with another operator's train cannot win; pairs whose names share no word need at least five trains.
- Result: 1,016 stations matched (977 by timetable, 39 by name as fallback); all key stations are correct.

5. Models and algorithms

No deep learning and no language model is used in the product.

Component	Type	Trained?
Delay model	8 LightGBM quantile-regression models (5, 10, 25, 50, 75, 90, 95, 98 %)	Yes, on historical delays
Cancellation model	Historical rate per line × station, pulled towards the line and product average when data is sparse	Computed from data
Baselines	Historical quantiles per line × station; historical median	Computed from data
Router	RAPTOR (Delling et al., 2012), own Python implementation	No – algorithm
Simulator	Monte Carlo (NumPy), samples from the predicted quantiles	No – uses the model

How the delay model was trained

Aspect	Choice
Unit of data	One row per train arrival or departure at a station
Features (10)	Product (RE/RB/S/long distance), operator, line, station, hour, weekday, stop number and relative position along the run, arrival/departure, delay at the previous stop
Target	Delay in minutes, clipped to −5...120; cancelled events handled separately
Split (by time)	Training March–June 2026: 2,713,317 events. Test July–August 2026: 578,578 events. Never a random split.
Previous-stop delay	Hidden for half of the training rows, so the model works both when planning ahead (unknown) and with live data (known)
Settings	Quantile objective, learning rate 0.08, 63 leaves, ≥ 200 rows per leaf, 300 rounds, 90 % feature sampling, 80 % bagging, seed 42, native categorical features
Most important	Station, line, operator, stop number, previous delay, hour
Safety rule	A new model is activated only if it scores at least as well as the current one
Output	Model v1, about 28 MB

6. System architecture

- **Offline (laptop):** download and filter delay data, train the model, run the replay evaluation. The large data never leaves the laptop.
- **Server:** FastAPI backend with the in-memory timetable, delay model and simulator; PostgreSQL, Redis, Celery jobs (daily data update, nightly accuracy check, weekly retrain); Caddy for HTTPS; Prometheus and Grafana for monitoring.
- **Frontend:** React single-page app (phone first, light and dark mode) with a Leaflet map.
- **Desktop:** the same backend and frontend packaged as one Windows program that opens in its own window.

7. Results

Delay model on the test months

Method	Average quantile loss (lower is better)
LightGBM, planning mode	1.006
LightGBM, previous-stop delay known	0.986
Historical quantiles per line × station	1.053
Historical median	1.786

Calibration of the upper range: 88.5 % of delays fell below the predicted 90th percentile, 93.9 % below the 95th and 97.3 % below the 98th. Cancellations are only marginally better than a single global rate (Brier 0.0338 vs. 0.0343).

Full journeys: historical replay

910 evening journeys were planned on every second day of July–August 2026 (12 origin/destination pairs, 4 departure times); 891 had complete real data to compare against.

Brier score (lower is better)	Timetable only	Historical median	Heimkommen
All connections work (happened 76 %)	0.241	0.168	0.127
Arrive ≤ 5 min late (happened 53 %)	0.473	0.391	0.216
Stranded (happened 4 %)	0.043	0.043	0.033

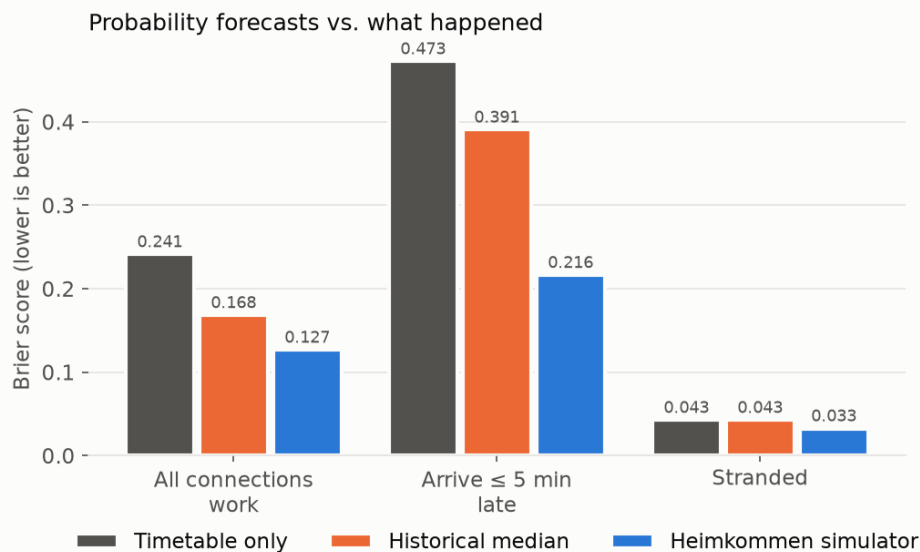


Figure 2 – Brier scores of the three methods in the replay.

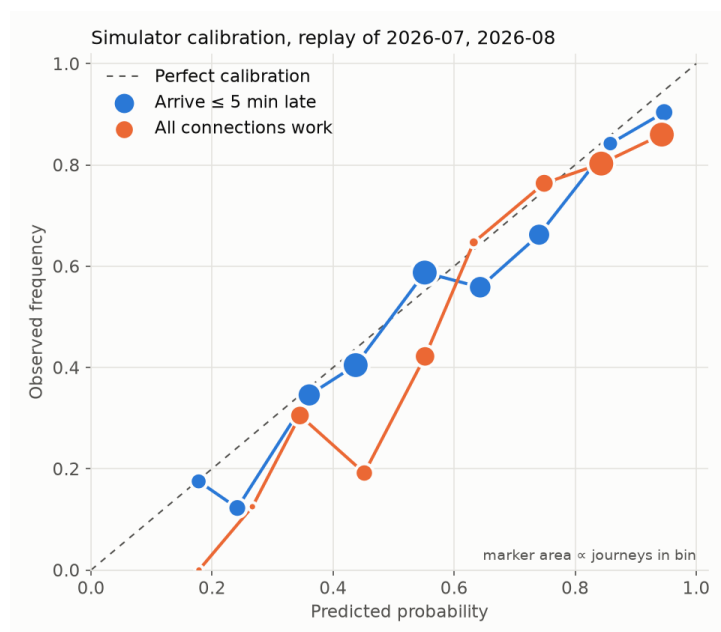


Figure 3 – Calibration: predicted probability vs. how often it actually happened.

Where it is weaker

Shortest change in the journey	Journeys	Predicted to work	Actually worked
≤ 5 min	93	38 %	27 %
6–10 min	187	66 %	59 %
> 10 min	240	86 %	80 %

The simulator is optimistic about tight connections. Train delays are simulated independently of each other, whereas in reality a bad evening affects many trains at once.

Live check

On 22 September 2026, 30 logged predictions were matched with what really happened: connection Brier score 0.139 vs. 0.207 for timetable-only.

A finding for travellers

From → To (Friday, Deutschlandticket)	Last departure	Chance to get home	Latest ≥ 95 % safe
Stuttgart Hbf → Villingen	22:17	81 %	21:14
Stuttgart Hbf → Donaueschingen	20:23	79 %	19:14
Stuttgart Hbf → Bad Dürkheim	20:23	18 %	none
Freiburg Hbf → Villingen	22:42	78 %	21:00

8. Software and libraries

Area	Libraries and tools (version)
Web API	FastAPI 0.141, Starlette 1.7, Uvicorn 0.53, Pydantic 2.13, pydantic-settings 2.15
Database	SQLAlchemy 2.0.54, Alembic 1.20, SQLite (local) / PostgreSQL 16 via psycopg 3.3
Data and ML	NumPy 2.5, pandas 3.0, PyArrow 25, LightGBM 4.7, SciPy 1.18
Background jobs	Celery 5.6, Redis 7 (client 8.1)
Security	bcrypt 5.0, python-jose 3.5, defusedxml 0.7, email-validator 2.3
Other backend	httpx 0.28, prometheus-client 0.26, tzdata
Frontend	React 18.3, React Router 6.30, TypeScript 5.9, Vite 5.4, Leaflet 1.9, openapi-typescript 7.13, ESLint 9.39
Analysis	Jupyter notebooks (nbclient), matplotlib 3.11
Testing and quality	pytest 9.1 (38 unit/API + 2 integration tests), testcontainers, Playwright 1.63, ruff 0.16, mypy 2.3
Infrastructure	Docker and Docker Compose, Caddy, Prometheus, Grafana, GitHub Actions
Desktop packaging	PyInstaller 6.22, pywebview 6.2 with Edge WebView2, Inno Setup 6, Pillow 12.3
Development tools	Python 3.14 (Docker images use 3.12), Node 26, Git, GitHub CLI, VS Code, Claude Code (AI coding assistant)

9. Quality and delivery

- 38 unit and API tests pass; ruff and mypy report no issues; the frontend lints and builds.
- Continuous integration on GitHub is green, including a job that runs the whole stack in Docker.
- The user interface was checked in a real browser at phone and desktop width, light and dark mode.
- The Windows app was tested end to end: install, launch, correct results, uninstall. The installer is 86 MB and needs no administrator rights.

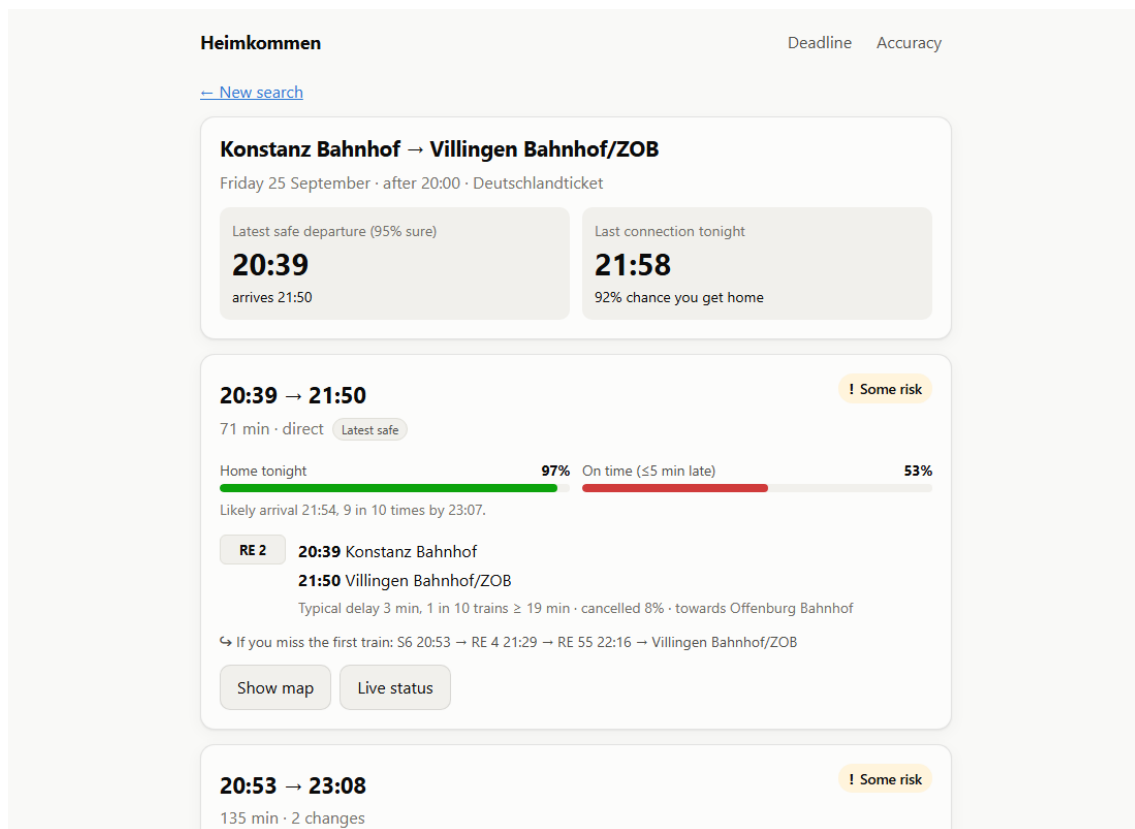


Figure 4 – The Windows desktop app: no login, same results as the web version.

10. Known limitations and next steps

Limitation	Next step
Long-distance trains mostly missing (the NVBW feed has only 4 such lines)	Add DB's national long-distance feed
Tight connections are predicted too optimistically	Model correlated delays (network-wide bad evenings), station-specific change times
Buses and trams assumed on time; only in the three region districts	Add bus delay data if it becomes available
Timetable valid until 12 December 2026	Import the next NVBW feed and rebuild
Cancellations barely better than a global rate	Use disruption and construction information
Windows installer is not code-signed	Sign it to avoid the SmartScreen warning

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